Adrià Mitjavila Ventura

Bioinformatician

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Bioinformatician with training in Molecular Genetics and Biotechnology and over 5 years of experience in omics data analysis, mainly transcriptomics and epigenomics. I am currently interested in regulation of gene expression. Outside of academia, I have participated in several scientific outreach events. On the other hand, I have practiced rowing at international level, which has given me the opportunity to apply the values acquired during training and competition, such as teamwork and perseverance, to my personal and professional life.

Education

PhD in Bioinformatics, *Universitat Autònoma de Barcelona (UAB)*, Online — *2022 - Present*

MSc. Bioinformatics and Biostatistics, *Universitat Oberta de Catalunya (UOC)*, Online — *2018 - 2022*

MSc. Molecular Genetics and Biotechnology, *Universidad de Sevilla (US)*, Seville (Spain) — *2017 - 2018*

BSc. Biotechnology, *Universitat de Girona (UdG)*, Girona (Spain) — *2012 - 2017*

Employment

PhD candidate in Bioinformatics — *11/2022 - Present*

Institut Josep Carreras (IJC) & *Institut Germans Trias i Pujol (IGTP)* — Badalona (Spain)

- Study of the variation Piwi-interacting RNAs (piRNAs) across closely-related mouse strains and species.
- Analysis and integration of small RNA-seq, RNA-seq, ChIP-seq, MNase-seq and other genomics data.
- Analysis of single-cell and spatial transcriptomics data (10x Xenium), including design of custom probes.
- Collaboration with external researchers and mentoring of undergraduate students.
- Outreach workshops addressed to high-school students.

Bioinformatician — *08/2021 - 11/2022*

Institut Josep Carreras (IJC) — Badalona (Spain)

- Study of the variation Piwi-interacting RNAs (piRNAs) across closely-related species.
- Analysis and integration of small RNA-seq, RNA-seq, ChIP-seq, MNase-seq and other genomics data.
- Collaboration with external researchers, and mentoring of BSc and MSc degree students.
- Outreach workshops addressed to high-school students.
- Master's thesis (UOC, 2022): *Analysis of the variation in piRNA expression across three* Mus *species*. Grade: 9.7/10 (**with honours**). Supervisor: Dr. Tanya Vavouri.

Bioinformatician — *07/2019 - 08/2021*

Istituto Europeo di Oncologia (IEO) — Milan (Italy)

- Study of several signaling pathways in the development of cancer and stem cell differentiation.
- Analysis and integration of RNA-seq, ChIP-seq and ATAC-seq data in the context of colorectal cancer.
- Comparison of distinct peak callers for ATAC-seq data.
- Comparison of sequencing depths for RNA-seq and their impact on differential expression analysis.

Master's thesis student & Research assistant — *09/2017 - 10/2018*

Instituto de Biomedicina de Sevilla (IBiS) — Seville (Spain)

- Study of Ccr4 and Xrn1 and their influence in nucleosome positioning and transcription in yeast.
- Computational analysis of MNase-seq data.
- MNase-qPCR experiments: DNA digestion, primer design, qPCR.
- Master's thesis (US, 2018): *Influence of Xrn1 in nucleosome positioning across* Saccharomyces cerevisiae *genome and its effects in transcription*. Grade: 9.2/10. Supervisors: Dr. Sebastián Chávez de Diego and Dr. Gabriel Gutiérrez Pozo.

Internship student — *07/2017 - 09/2017*

Institut d'Investigació Biomèdica de Girona (IdIBGi) — Girona (Spain)

- Identification of new factors (genetic and bacterial) related to adipose tissue physiology and obesity.
- DNA and RNA extraction from animal and human biopsies/feces.

Internship & Bachelor's thesis student — *09/2016 - 07/2017*

Microbial. Sistemes i aplicacions analítiques, S.L. — Girona (Spain)

- Analysis of the mucosa-associated microbiota from human intestinal biopsies
- Analysis of food and water samples for pathogen identification and food forensics.

- DNA extraction, qPCR, molecular clonning, analysis, statistical analyisis of NGS (metagenomics) data.
- Bachelor's thesis (UdG, 2017): *Butyrate-induced changes in the diversity of the intestinal mucosa-associated microbiota in colorectal cancer patients submitted to a lateral ileostomy*. Grade: 9.4/10. Supervisor: Prof. Jesús García-Gil.

Publications

Journal Articles

Casas, Mitjavila-Ventura *et al*. **Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse**. *EMBO Journal*. 2025. DOI: 10.1038/s44318-025-00475-4.

Begley *et al*. **Xrn1 influence on gene transcription results from the combination of general effects on elongating RNA pol II and gene-specific chromatin configuration**. *RNA Biology*. 2021. DOI: 10.1080/15476286.2020.1845504.

Presentations and posters

Mitjavila-Ventura, Casas et al. **Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse**. Poster presentation at the *Jornades de Bioinformàtica i Genòmica* from the Catalan Society of Biology (Lleida, December 2024).

Mitjavila-Ventura & Vavouri. **Transposons and the evolution of piRNA cluster expression in mice**. Poster presentation at the piRNAs and PIWI proteins meeting (Montpellier, April 2024).

Skills

Languages

- Catalan (*native*), Spanish (*bilingual*), English (*IELTS 7.5 - C1*), Italian (*basic*)

Bioinformatics

- *Omics data analysis*: Microarray, (small) RNA-seq, ChIP-seq, ATAC-seq, MNase-seq
- *Genomic Browsers*: IGV, UCSC Genome Browser, ENSEMBL Browser (multiple genomes)
- *Other*: Analysis of sequences, ENSEMBL Compara, Biomart, BLAST

Informatics

- *Languages*: R/Bioconductor, Bash, AWK, Python (*basic*)
- *Other*: R Markdown, Snakemake, Conda, Docker, Singularity, Git, HPC (PBS/Slurm), LaTeX

Development

R packages

- `plOtMICS`: visualization of omics and sequencing data in R.
- `ggmitji`: cool functions to complement ggplots.
- `chromHMMviewR`: visualize outputs from `ChromHMM`.

See more on my Github profile

Organizations

Catalan Society of Biology (SCB). 2018 - Present

Biotechnologists Association of Catalunya (ASBTEC). 2017 - 2024

- *Active collaborator from 2018 to 2024*
- *Board member and communication manager from 2019 to 2022*

Honours & Awards

INPhINIT Incoming PhD Fellowship (*La Caixa* Foundation, 2022-2025)

Master's thesis **with honours** (UOC, 2022)

SEMM PhD Fellowship (SEMM, 2019, *declined*)

Elite athlete (BOE, 2015-2021)

World Rowing Under-23 Champion (FISA, 2016)